

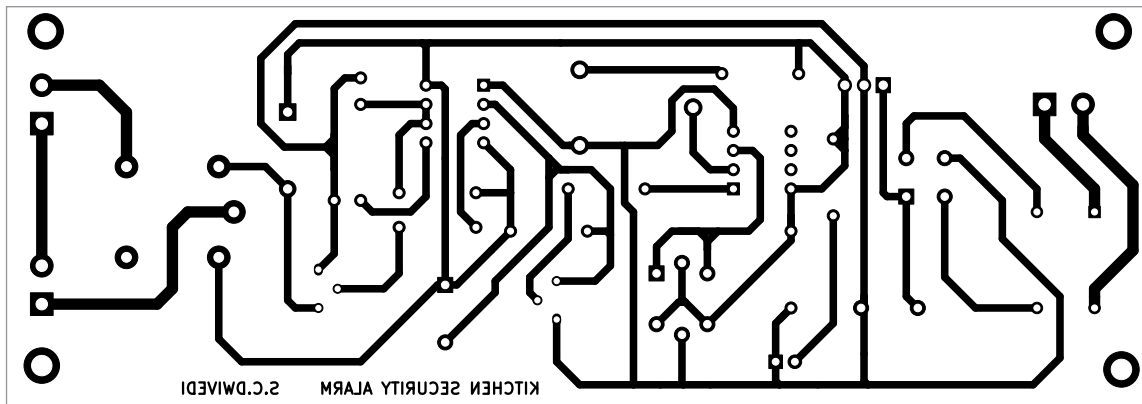


Fig. 2: Actual-size PCB of kitchen security alarm circuit

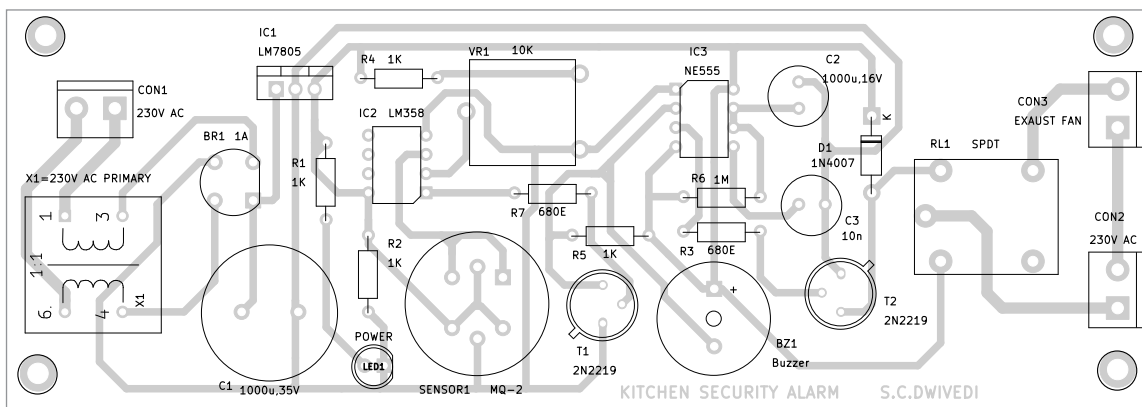


Fig. 3: Component layout of the PCB

buzzer is meant to raise an alarm.

Working of the circuit is simple. After connecting 230V AC across CON1, when power is switched on, LED1 glows to indicate that the circuit is ready to use. In case of smoke or gas leakage beyond the set limit, the MQ2 sensor's output goes higher at pin 3 of IC LM358 than the reference voltage at its pin 2. Thus, the output of op-amp IC LM358 at its pin 1 goes high. As a result, transistor T1 conducts and the buzzer sounds to indicate smoke or gas leakage.

Also, at the same time, IC NE555 triggers and transistor T2 conducts to energise relay RL1. If an exhaust fan is connected between the common and normally closed contacts of the relay, it will switch on and keep operating for around 18 minutes because of timer IC NE555 that has been wired in monostable multivibrator mode.

Here, the NE555 timer IC has been wired in monostable multivibrator mode. Based on the above-mentioned timing components resistor R6 and capacitor C2, the on time output of IC3 will be around 18 minutes.

This output pulse of NE555 stays high based on its pulse width, which sets the pre-defined time period. This time period can be set by selecting the values of resistor R6 and capacitor C2 using the relationship:

$$\text{Output Pulse Width (secs)} = 1.1 \times R6 \times C2$$

In this circuit, the value of R6 = 1M and value of C2 = 1000μF. So, if we calculate the time in seconds, it will be around 1100 seconds, which is roughly 18 minutes.

As we are using MQ2 sensor to compare the output voltage of LM358 with the reference voltage, to calibrate the sensor, set the reference

voltage to around 0.5V using pot VR1. Then create some smoke or LPG gas leakage near the sensor. After some time, the output at pin 3 of op-amp LM358 would go higher than the reference voltage at pin 2. This will make the output at pin 1 of LM358 to trigger the buzzer. If the buzzer sounds, it means the calibration is successful. If the buzzer does not

sound, adjust pot VR1.

Construction and testing

An actual-size, single-side PCB layout for the security alarm is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable box. Fix LED1 and CON2 in front of box and connect exhaust fan on the rear side of the box.

After assembling the circuit, install the unit not too far from the gas stove, so that when gas leaks or there is excessive smoke in the area, it can be easily accessed by the sensor. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-kitchen-security-alarm-using-mq2-sensor>

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